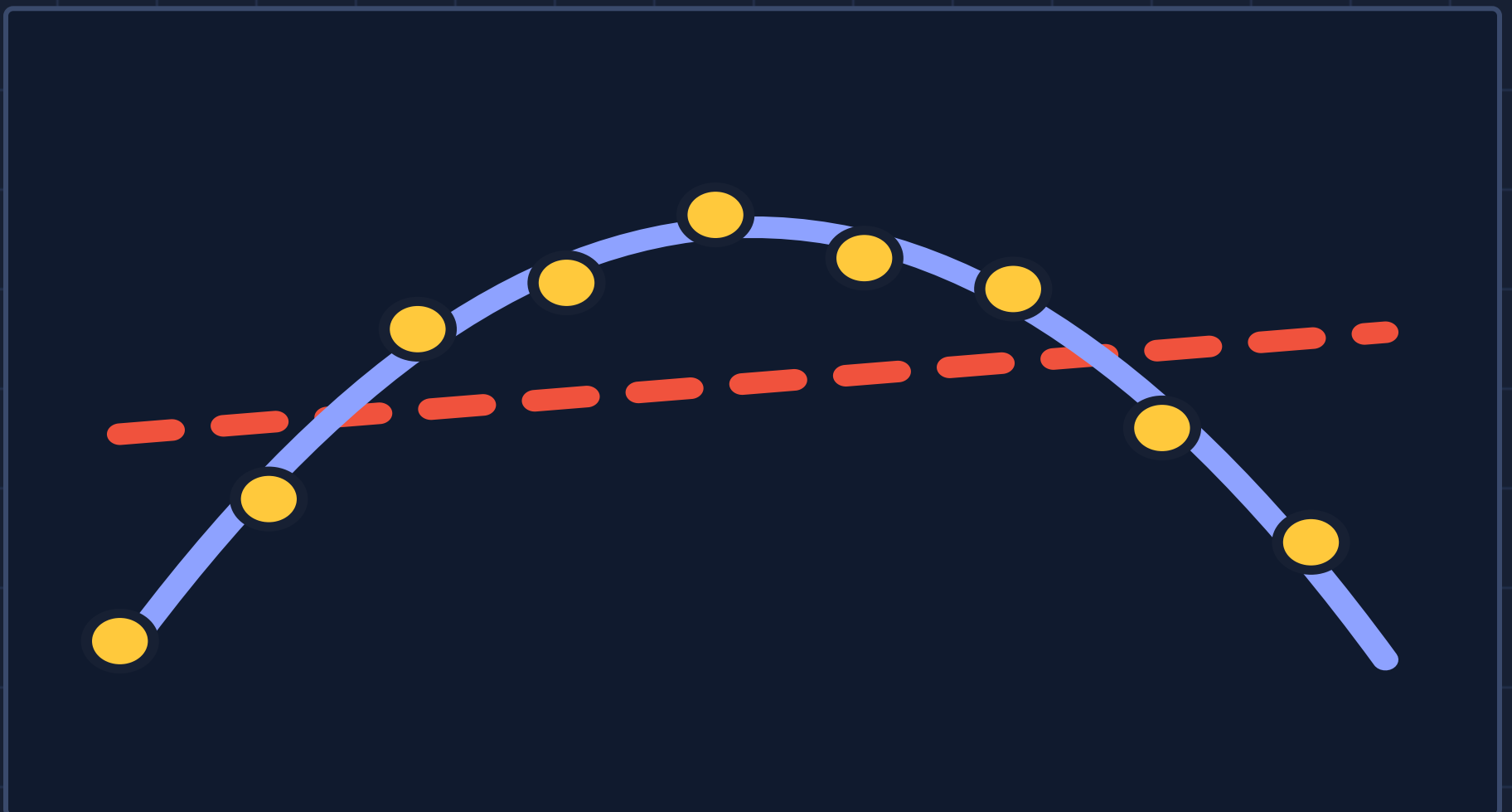


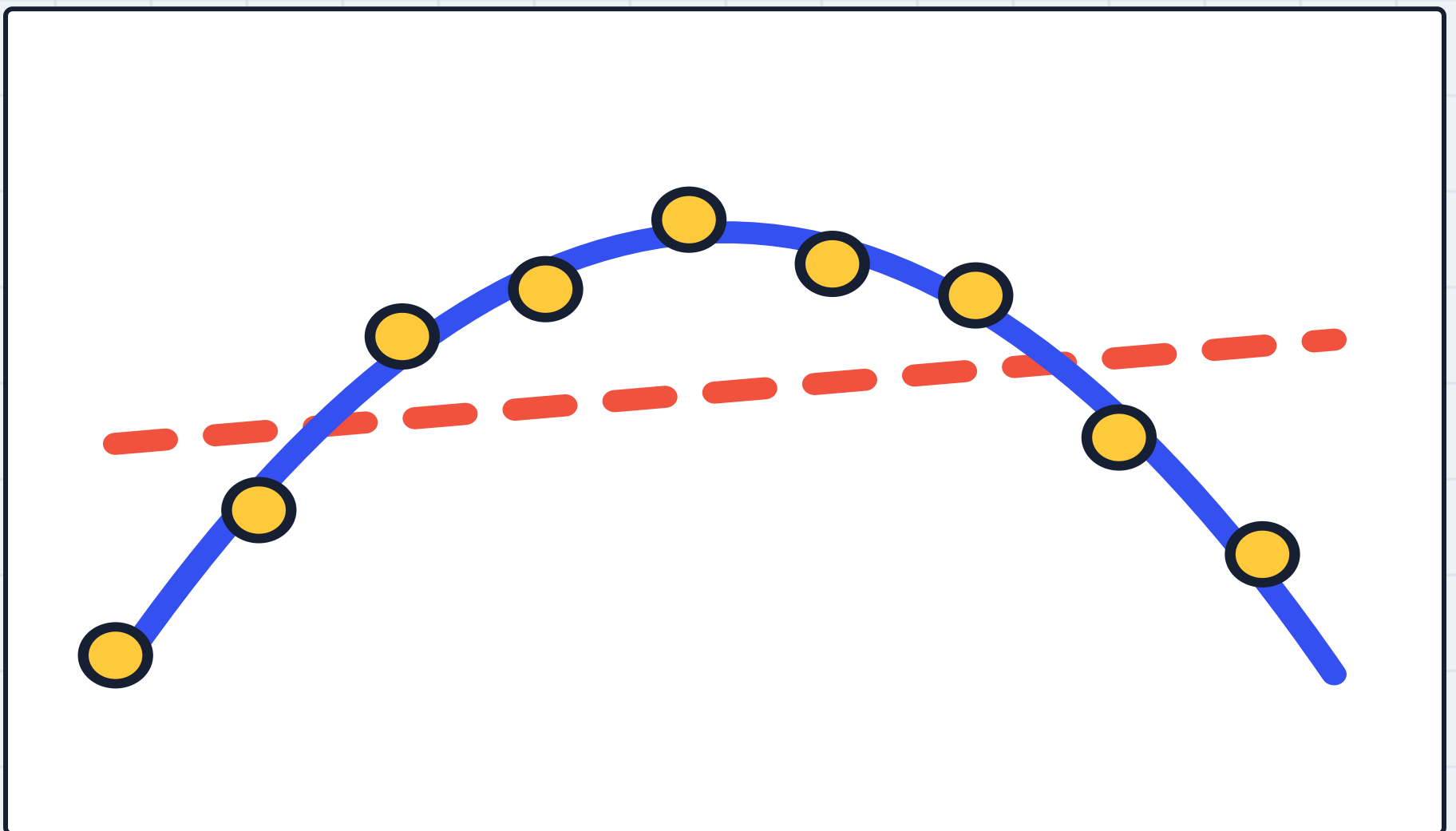
When a straight line isn't enough

Polynomial regression in plain words. No code, only the idea.



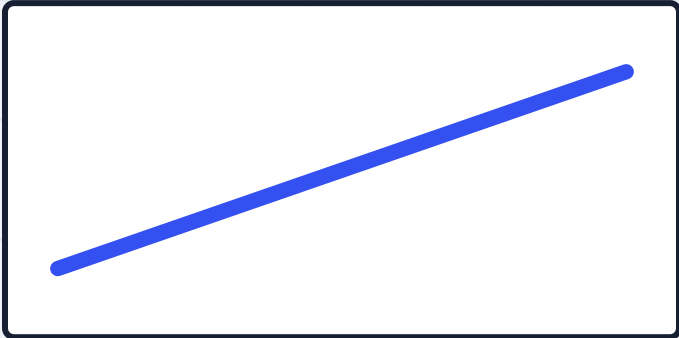
Real data bends

Think of a car braking. Drive faster and the car needs much more road, not just a little more. Plot it and you get a curve.



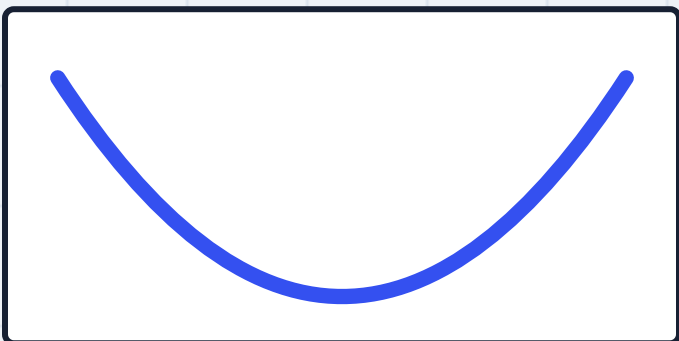
Red dashes: a straight line ignores the bend. **Blue:** a curve follows it.

The trick: add powers of **x**



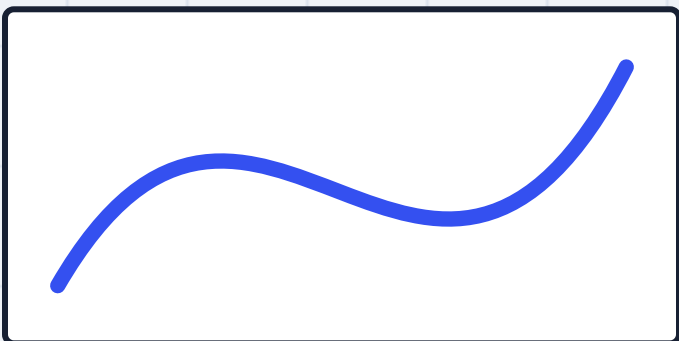
$$y = a + b \cdot x$$

A straight line. No bends.



$$y = a + b \cdot x + c \cdot x^2$$

Degree 2. One smooth curve.



$$y = a + b \cdot x + c \cdot x^2 + d \cdot x^3$$

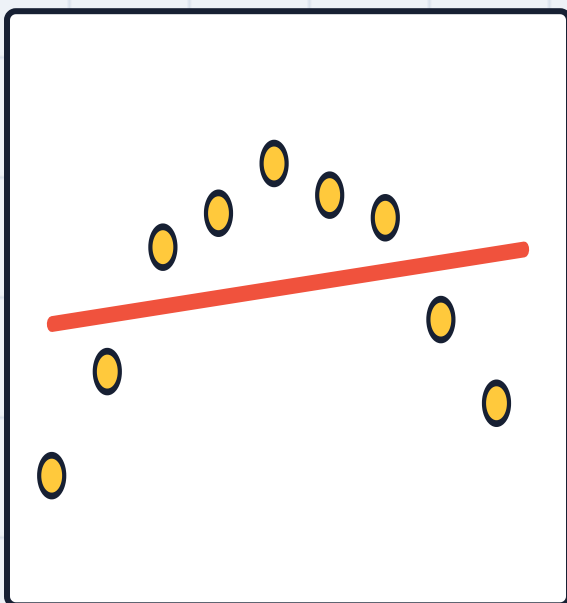
Degree 3. An S-shaped wiggle.

x^2 and x^3 are just extra columns of data.

Underneath, it is still linear regression. The degree is how flexible the curve gets.

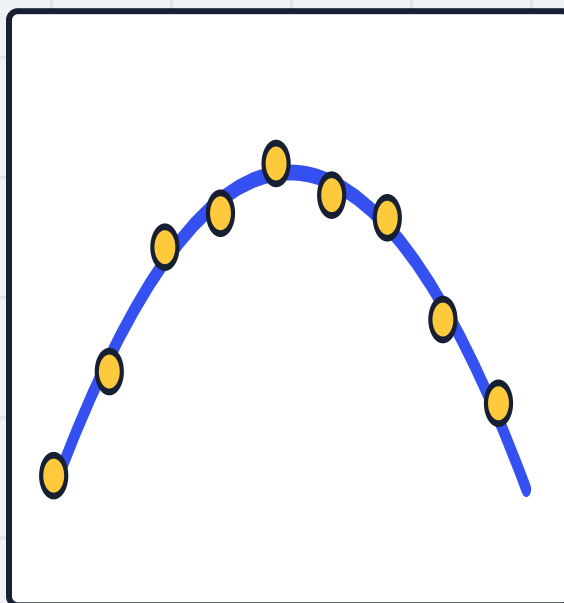
Too simple, just right, too eager

Same dots, three models. Only one of them learned the pattern.



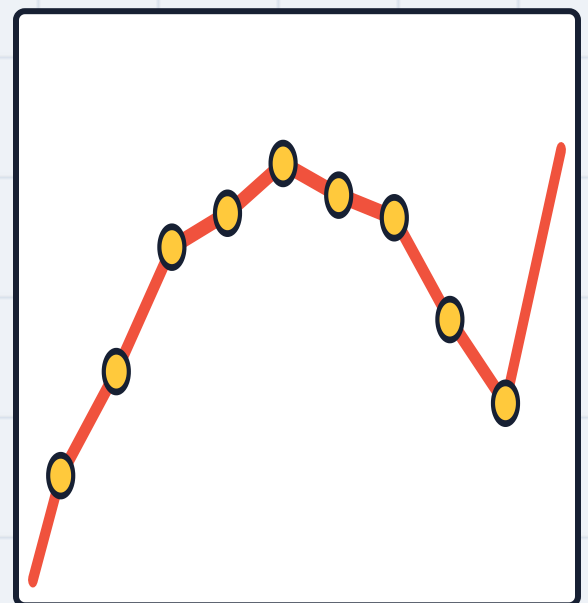
Underfit

Too simple. Misses the pattern, so it's wrong on old and new data.



Just right

Follows the trend and ignores the random noise.



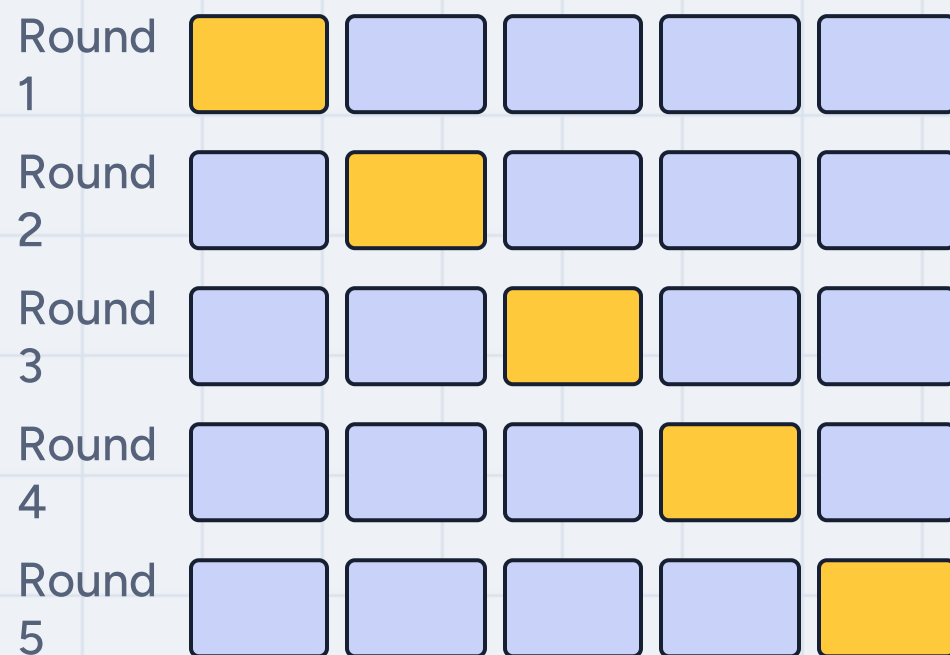
Overfit

Touches every dot. It memorized the noise and fails on new data.

The trap: an overfit model looks perfect on the data it trained on.

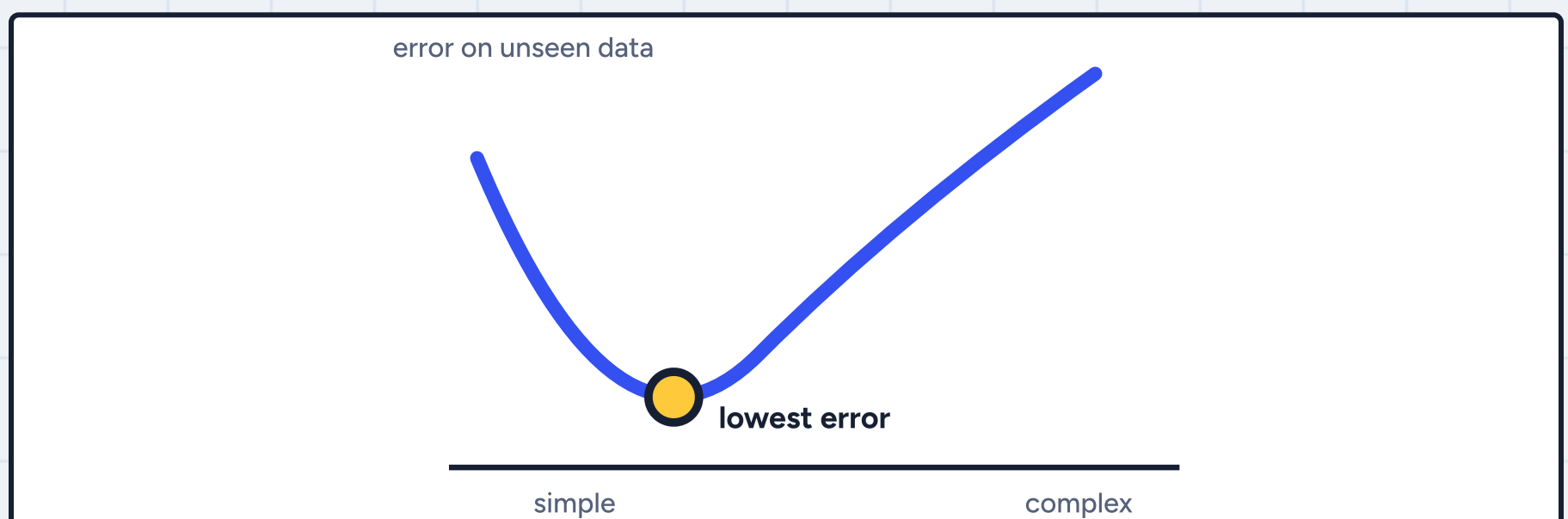
How to pick the degree

Quiz the model on data it has never seen. This is called cross-validation.



Cut the data into 5 parts. Train on 4, test on the 1 left out. Rotate until each part has been the test once, then average the errors.

Yellow is the test part. Blue is for training.

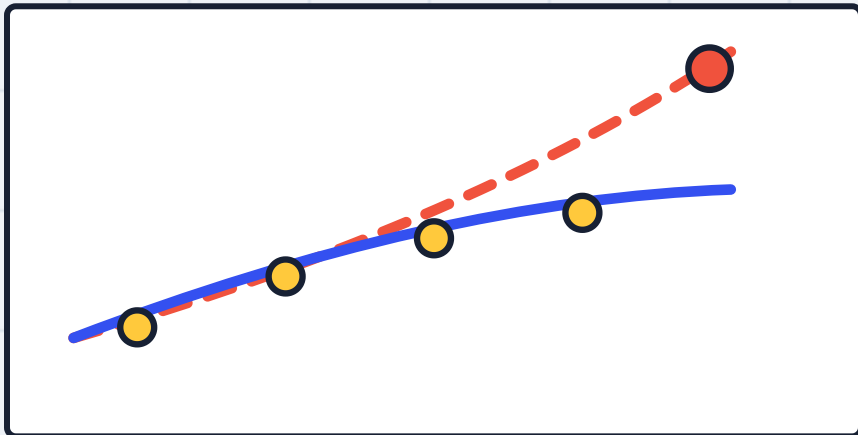


Repeat for each degree. If two tie, pick the lower one. Simple wins.

Watch out, and other options

One bad point can bend everything

The model squares every error, so a far-away point gets a huge say. Higher degrees chase it even harder.



Plot your data first. Remove real mistakes. If the odd points are genuine, use RANSAC or Huber regression.

Splines

Many small curves joined smoothly at points called knots. Good when the data acts differently in different regions.

SVR

Support Vector Regression ignores points that sit inside a safe margin. Good when outliers can't be removed.

Both give more control over the fit, but are harder to explain.

What to remember

- Real data often curves. A straight line misses it.
- Add x^2 or x^3 and the same model can bend.
- Start with degree 2 or 3. Let cross-validation confirm it.
- Too low underfits. Too high memorizes noise.
- Watch for outliers. Try splines or SVR if the fit gets shaky.

Based on the DataCamp tutorial "Polynomial Regression" by Dario Radečić.